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WAFER SAMPLE SEGMENTATION APPARATUS

Abstract

A wafer sample segmentation apparatus is disclosed and the wafer sample segmentation apparatus according to some embodiments comprises a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part facing the other surface facing one surface of the sample and capable of pressing a pair of pressing regions positioned on both sides of one region in the sample by being moved toward the sample; an indentation part facing the other surface of the sample and capable of moving toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample; and a measuring part capable of measuring a pressing force acting on the sample along the pressing direction of the pressing part.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit under 35 U.S.C. § 119(a)-(d) of Korean Patent Application No. 10-2024-0034062 filed on Mar. 11, 2024, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a wafer sample segmentation apparatus for manufacturing a specimen by segmenting a wafer sample.

BACKGROUND

[0003] A semiconductor manufacturing process may include various processes, and in order to determine whether results of the various processes are normal, a cross-section of a specimen manufactured from a wafer that has gone through the process may be inspected.

[0004] The cross-section of the specimen may be inspected in any of a variety of ways, such as by taking high-magnification images. For measuring the uniformity of the microstructure of the specimen, it may be advantageous for the cross-section of the fabricated specimen to have high shape density.

SUMMARY

[0005] The embodiments seek to provide a wafer sample segmentation apparatus that may improve the segmentation density and the cross-section flatness.

[0006] The embodiments seek to provide a wafer sample segmentation apparatus that may effectively control and keep constant a pressing force acting on the sample.

[0007] The embodiments seek to provide a wafer sample segmentation apparatus that may uniformly provide the pressing force to the sample.

[0008] The embodiments seek to provide a wafer sample segmentation apparatus that may adjust a position relationship between components for the segmentation.

[0009] The embodiments seek to provide a wafer sample segmentation apparatus that may align the crystal direction of the sample and the blade direction of the indentation part.

[0010] A wafer sample segmentation apparatus according to an embodiment includes a supporting part, a pressing part, an indentation part, and a measuring part.

[0011] The supporting part faces the sample one surface of the wafer and supports one region of the sample. The pressing part faces the other surface facing one surface of the sample and is moved toward the sample to press a pair of pressing regions positioned on both sides of the one region in the sample.

[0012] The indentation part faces the other surface of the sample and moves toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample. The measuring part measures a pressing force acting on the sample along the pressing direction of the pressing part.

[0013] On the other hand, in a wafer sample segmentation apparatus according to an embodiment, the supporting part may support one region of a sample while facing one surface of the sample of a wafer, and the pressing part may face the other surface opposite to one surface of the sample and move toward the sample to press the pressing region of the sample.

[0014] The indentation part may face the other surface of the sample and move toward the sample to indent the segmentation region of the sample; and the measuring part may measure a pressing force acting on the sample along the pressing direction of the pressing part,

[0015] If the measurement value of the measuring part falls within a pressing reference range, the indentation part may be moved toward the sample to indent the segmentation region.

[0016] On the other hand, in a wafer sample segmentation apparatus according to an embodiment,

the supporting part may support one region of a sample while facing one surface of the sample of a wafer. The pressing part may be provided as a pair to each face the other surface of one surface of the sample, and move toward the sample to press a pair of pressing regions positioned on both sides of the one region in the sample.

[0017] The indentation part may face the other surface of the sample and move toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample, wherein the supporting part may be positioned between the pair of pressing parts with reference to the arrangement direction of the pair of pressing parts, and the pair of pressing parts may be movable along the arrangement direction.

[0018] Embodiments may provide the wafer sample segmentation apparatus that may improve a segmentation accuracy and a flatness of a cross-section.

[0019] Embodiments may provide the wafer sample segmentation apparatus that may effectively control and keep constant a pressing force acting on the sample.

[0020] Embodiments may provide the wafer sample segmentation apparatus that may uniformly provide the pressing force to the sample.

[0021] Embodiments may provide the wafer sample segmentation apparatus that may adjust position relationships between components for the segmentation.

[0022] Embodiments may provide the wafer sample segmentation apparatus that may align the crystal direction of the sample and the blade direction of the indentation part.

[0023] According to some aspects, a wafer sample segmentation apparatus is generally described. In some embodiments, the apparatus comprises a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part facing the other surface facing one surface of the sample and capable of pressing a pair of pressing regions positioned on both sides of the one region of the sample by being moved toward the sample; an indentation part facing the other surface of the sample and capable of moving toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample; and a measuring part capable of measuring a pressing force acting on the sample along the pressing direction of the pressing part.

[0024] In some embodiments, the apparatus comprises: a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part facing the other surface opposite to the one surface of the sample and capable of moving toward the sample to press the pressing region of the sample; an indentation part facing the other surface of the sample and capable of moving toward the sample to indent the segmentation region of the sample; and a measuring part capable of measuring a measurement value acting on the sample along the pressing direction of the pressing part, wherein: if the measurement value of the measuring part falls within a pressing reference range, the indentation part is capable of moving toward the sample to indent the segmentation region.

[0025] In some embodiments, the apparatus comprises: a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part provided as a pair facing the other surface opposite to the one surface of the sample of the sample, and capable of moving toward the sample to press a pair of pressing regions positioned on both sides of the one region in the sample; and an indentation part facing the other surface of the sample and capable of moving toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample, wherein the supporting part is positioned between the pair of pressing parts with reference to an arrangement direction of the pair of pressing parts, and the pair of pressing parts is movable along the arrangement direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. **1** is a view showing a wafer sample segmentation apparatus according to some embodiments.

[0027] FIG. **2** is a view showing a sample moving part of a wafer sample segmentation apparatus according to some embodiments.

[0028] FIG. **3** is a view showing a sample segmentation part of a wafer sample segmentation apparatus according to some embodiments.

[0029] FIG. **4** is a view showing a sample being placed in a sample arrangement part, according to some embodiments.

[0030] FIG. **5** is a view showing a supporting part, a pressing part, and an indentation part, according to some embodiments.

[0031] FIG. **6** is a view showing an arrangement state in which a sample is placed in a sample arrangement part during a sample segmentation process, according to some embodiments.

[0032] FIG. **7** is a view showing a supporting state in which a sample is supported by a supporting part during a segmentation process of a sample, according to some embodiments.

[0033] FIG. **8** is a view showing a pressing state in which a sample is pressed by a pressing part during a sample segmentation process, according to some embodiments.

[0034] FIG. **9** is a view showing an indentation state in which a sample is indented by an indentation part during a sample segmentation process, according to some embodiments.

[0035] FIG. **10** is a view showing a segmentation state in which a sample is segmented and a specimen is produced during a sample segmentation process, according to some embodiments.

[0036] FIG. **11** is a view showing a support part provided in a sample segmentation part, according to some embodiments.

[0037] FIG. **12** is a view showing a pressing part provided in a sample segmentation part, according to some embodiments.

[0038] FIG. **13** is a view showing an indentation part provided in a sample segmentation part, according to some embodiments.

[0039] FIG. **14** is a view showing a sample arrangement part provided in a sample segmentation part, according to some embodiments.

[0040] FIG. **15** is a view showing a camera provided in a sample segmentation part, according to some embodiments.

[0041] FIG. **16** is a view showing an alignment process of a pressing part and an indentation part, according to some embodiments.

[0042] FIG. **17** is a view showing an alignment process between a crystal direction of a sample and a blade direction of an indentation part, according to some embodiments.

[0043] FIG. **18** is a block diagram showing a control relationship between components through a controller, according to some embodiments.

DETAILED DESCRIPTION

[0044] In the following detailed description, only certain embodiments of the present disclosure have been shown and described, simply by way of illustration.

[0045] As those skilled in the art would realize, the described embodiments may be modified in various different ways, all without departing from the spirit or scope of the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature and not restrictive. Like reference numerals designate like elements throughout the specification.

[0046] In the present specification, duplicate descriptions for the same components are omitted.

[0047] Also, in present specification, it is to be understood that when one component is referred to as being “connected” or “coupled” to another component, it may be connected or coupled directly to the other component or be connected or coupled to another component with the other component intervening therebetween. On the other hand, in this specification, it is to be understood that when

one component is referred to as being “connected or coupled directly” to another component, it may be connected or coupled to the other component without another component intervening therebetween.

[0048] It is also to be understood that the terminology used herein is only for the purpose of describing particular embodiments, and is not intended to be limiting of the disclosure.

[0049] Singular forms are to include plural forms unless the context clearly indicates otherwise.

[0050] It will be further understood that term “comprises” or “have” used in the present specification specify the presence of stated features, numerals, steps, operations, components, parts, or a combination thereof, but does not preclude the presence or addition of one or more other features, numerals, steps, operations, components, parts, or a combination thereof.

[0051] Also, as used herein, the term “and/or” includes any plurality of combinations of items or any of a plurality of listed items. In this specification, “A or B” may include “A”, “B”, or “A and B”.

[0052] The specimen of the wafer may be produced through a segmentation method. The segmentation method may be useful for producing the specimen with the cross-section created along the crystal direction of the wafer, and the specimen with the cross-section may be produced by segmenting the wafer or the sample prepared by subdividing from the wafer.

[0053] Generally, the smoother the cross-section of the specimen produced using the segmentation method, the better the inspection accuracy. To smoothly create the cross-section of the specimen, a method of splitting the sample by indenting the sample provided with a bending force may be used.

[0054] In the segmentation method, the smoothness of the cross-section depends on the magnitude and uniformity of the bending force acting on the sample, and may vary depending on the angle between the blade direction defined in the indentation tool and the crystal direction of the sample.

[0055] Therefore, by producing the specimen via segmenting the wafer sample, stably controlling the bending force acting on the specimen, providing a uniform bending force to the sample using the cross-section as a reference, and effectively aligning the crystal direction of the sample and the blade direction of the indentation tool are relatively important tasks in this field of technology.

[0056] FIG. 1 shows a wafer sample segmentation apparatus **1** according to some embodiments. The wafer sample segmentation apparatus **1** includes a sample moving part **20** that moves a sample **10** (See FIG. 2) of the wafer and a sample segmentation part **30** that segments the sample **10** received through the sample moving part **20**.

[0057] Various processes may be performed in a semiconductor manufacturing process used in the semiconductor industry. To determine the results of the various processes, a cross-section inspection of a semiconductor substrate, for example a wafer, may be performed.

[0058] In some embodiments, the sample **10** of the wafer may be a piece subdivided from the wafer, and if possible, the wafer itself may be used as the sample **10**. Hereinafter, for better understanding and ease of explanation, the sample **10** of the wafer is described as a piece divided (e.g., subdivided) from the wafer.

[0059] The sample moving part **20** may move the sample **10** toward the sample segmentation part **30** while the sample **10** is placed. The sample moving part **20** may be moved toward the sample segmentation part **30** as is, or a part of the sample moving part **20** may be moved toward the sample segmentation part **30**. If necessary, at least part of the sample segmentation part **30** may be moved toward the sample moving part **20** to receive the sample **10**.

[0060] The sample segmentation part **30** is prepared to segment the sample **10** transported from the sample moving part **20**. The sample segmentation part **30** may produce a specimen with a cross-section cut by segmenting the sample **10**. The cross-section may be extended in a direction across one surface **11** (see FIG. 6) of the sample **10** and may be created through the segmentation of the sample **10**.

[0061] The cross-section of the specimen may be inspected by taking pictures using high resolution or high magnification photography. For measuring the uniformity of the microstructure of the

specimen, it may be advantageous for the cross-section of the manufactured specimen to have high formation accuracy.

[0062] In other words, the smoother the cross-section of the specimen, the more advantageous it may be for high-precision inspection. Therefore, a specimen having a cross-section with high planarity and high accuracy may be more advantageous for the specimen inspection. Some embodiments seek to effectively produce the specimen including the cross-sections of the specimen with high formation accuracy.

[0063] FIG. 2 shows the sample moving part **20** of the wafer sample segmentation apparatus **1**. The sample moving part **20** may be provided together with a base part where the sample segmentation part **30** described above is placed, or the sample moving part **20** may be provided to have a separate base and be movable.

[0064] The sample moving part **20** may accommodate and move the sample **10** described above. The sample **10** may be divided from the wafer in advance before being accommodated in the sample moving part **20** and provided to the sample moving part **20**. However, if necessary, it is possible for the sample **10** to be subdivided while being accommodated in the sample moving part **20**.

[0065] FIG. 2 shows one sample moving part **20**, but the number of the sample moving parts **20** may vary depending on the need. Additionally, the way that the sample moving part **20** transports the sample **10** to the sample segmentation part **30** may also vary.

[0066] FIG. 3 shows the sample segmentation part **30** of the wafer sample segmentation apparatus **1**. The sample segmentation part **30** may produce the cross-section of the specimen by receiving the sample **10** through the sample moving part **20** and segmenting it.

[0067] The sample segmentation part **30** may include at least one among a sample arrangement part **400**, in which a sample **10** is arranged, a supporting part **100** supporting the sample **10** arranged in the sample arrangement part **400**, a pressing part **200** that generates a bending force on the sample **10** by pressing the sample **10** supported by the supporting part **100**, and an indentation part **300** that indents the sample **10** so that the sample **10** in which the bending force occurs is split.

[0068] FIG. 4 shows the shape in which the sample **10** is arranged in the sample arrangement portion **410** of the sample arrangement part **400**. In some embodiments, the sample arrangement part **400** includes a sample arrangement portion **410** in which the sample **10** is arranged. The sample **10** may be segmented while being arranged in the sample arrangement portion **410** to create a specimen.

[0069] The pressing part **200** may include a pressing portion **210** for pressing the sample **10**, and the indentation part **300** may include an indentation portion **310** for indenting the sample **10**. The sample arrangement portion **410** may be positioned toward one surface **11** of the sample **10**, and the pressing portion **210** and the indentation portion **310** may be positioned toward the other surface **12** (see FIG. 6) of the sample **10**.

[0070] The sample **10** may be positioned on the sample arrangement portion **410** and segmented by the pressing portion **210** and the indentation portion **310** to produce the specimen.

[0071] FIG. 5 shows the supporting portion **110**, the pressing portion **210**, and the indentation portion **310** with the above-mentioned sample arrangement portion **410** omitted in the drawing.

[0072] The supporting portion **110** faces one surface **11** of the sample **10** of the wafer and supports one region **13** (see FIG. 7) of the sample **10**. The one region **13** may be defined as the part supported by the supporting portion **110** in the sample **10**. The one region **13**, that is the support region, may be defined in any of a variety of parts of the sample **10**.

[0073] The supporting portion **110** may support the sample **10** by being in contact with the sample **10** in a form of a point, a line, or a surface. FIG. 5 shows that the supporting portion **110**, extended in the form of a bar or an arm, supports one surface **11** of the sample **10** in the form of a line or surface.

[0074] The pressing portion **210** faces the other surface **12** opposite to one surface **11** of the sample

10, and moves toward the sample **10** to press the pressing region **14** (see FIG. **8**) of the sample **10**. The pressing region **14** may refer to a portion pressed by the pressing portion **210** in the sample **10**. [0075] The position of the pressing region **14** relative to one region **13** of the aforementioned supporting portion **110** may vary. In some embodiments, the pressing region **14** may be spaced apart from the one region **13**, and may be defined as a pair and positioned on both sides of the one region **13** in the sample **10**. In some embodiments, the pressing portion **210** may be provided as a pair and arranged to be spaced apart on both sides of the supporting portion **110**.

[0076] The pressing portion **210** may move toward the other surface **12** of the sample **10**, such that each pressing portion **210** spaced apart on the other surface **12** of the sample **10** presses each of the pressing regions **14** of the sample **10**.

[0077] The pressing portion **210** may press the sample **10** by being in contact with the sample **10** in the form of a point, a line, or a surface. FIG. **5** shows the pressing portion **210**, extended in the form of a bar or an arm, pressing one surface **11** of the sample **10** in the form of a line or surface.

[0078] The other surface **12** of the sample **10** may be the opposite surface of one surface **11** of the sample **10**. That is, the pressing portion **210** may be positioned on the opposite side of the supporting portion **110** with the sample **10** as a reference and may press the sample **10** toward the supporting portion **110**.

[0079] The indentation portion **310** may face the other surface **12** of the sample **10** and be moved toward the sample **10** to indent the segmentation region **15** (see FIG. **9**) positioned between the pair of pressing regions **14** in the sample **10**.

[0080] The segmentation region **15** may be a region where the indentation occurs by being in contact with the indentation portion **310** in the sample **10**.

[0081] The segmentation region **15** may be defined between the aforementioned pair of pressing regions **14**.

[0082] That is, the indentation portion **310** may indent the other surface **12** of the sample **10** by being positioned between the pair of pressing units **210** and moved toward the sample **10**. When the indentation occurs in the sample **10**, which is pressed by the pressing portion **210**, a bending force is generated. Splitting may occur with the indentation mark as a starting point due to the concentration of the stress, and accordingly, a specimen having a cross-section crossing one surface **11** and the other surface **12** of the sample **10** may be produced.

[0083] In some embodiments, a first direction **D1**, a second direction **D2**, and a third direction **D3** may be defined for better understanding and ease of description. The first direction **D1** may be parallel to one surface **11** of the sample **10**, may be the length direction of the sample **10**, and may be a direction in which the pair of pressing portion **210** and the indentation portion **310** are aligned/arranged.

[0084] The second direction **D2** may be parallel to one surface **11** of the sample **10**, and may be defined to cross the first direction **D1**. For example, the second direction **D2** may be the width direction of the sample **10** and an elongated direction of the supporting portion **110**, the pressing portion **210**, and the indentation portion **310**. The second direction **D2** may be defined as substantially orthogonal to the first direction **D1**.

[0085] The third direction **D3** may be a direction passing through one surface **11** of sample **10**, and may be defined as a direction crossing the first direction **D1** and the second direction **D2**. The third direction **D3** may be the thickness direction of the sample **10** and the movement direction of the pressing portion **210** and the indentation portion **310**.

[0086] FIG. **6** to FIG. **10** show the process in which the sample **10** is segmented in the wafer sample segmentation apparatus **1** and a specimen is produced.

[0087] FIG. **6** shows the arrangement state **P1** of the sample **10** during the segmentation process of the sample **10**. In the arrangement state **P1**, the sample **10** may be arranged in the sample arrangement portion **410**, and the sample **10** may be supported by the sample arrangement portion **410**.

[0088] The sample arrangement portion **410** may be in a state that is contact with the sample **10**, and the pressing portion **210** and the indentation portion **310** may be in a state where the pressing portion **210** and the indentation portion **310** are spaced apart from sample **10**. The pressing portion **210** and the indentation portion **310** may be in a state where the pressing portion **210** and the indentation portion **310** are moved away along the third direction D3 from the other surface **12** of the sample **10**.

[0089] In the arrangement state P1, the alignment of the crystal direction D4 (see FIG. 14) of the sample **10**, the alignment of the pressing portion **210**, and the alignment of the indentation portion **310** may be performed. The detailed description of these alignment process will be provided later.

[0090] FIG. 7 shows a supporting state P2 of the sample **10** during the segmentation process of the sample **10**. In the supporting state P2, as the sample arrangement portion **410** moves in the third direction D3 away from one surface **11** of the sample **10**, one surface **11** of the sample **10** may be supported by the supporting portion **110**.

[0091] In some embodiments, the sample arrangement portion **410** may include an accommodation space **415** capable of accommodating the supporting portion **110**. In the aforementioned arrangement state P1, the supporting portion **110** may be accommodated in the accommodation space **415** of the sample arrangement portion **410** and may be spaced apart from one surface **11** of the sample **10**.

[0092] From the arrangement state P1, the sample arrangement portion **410** may be moved along the third direction D3 to be separated from one surface **11** of sample **10** to become the supporting state P2. By moving the sample arrangement portion **410**, the supporting portion **110** may be reflexively drawn out from the accommodation space **415**, and support one surface **11** of the sample **10**, and support one region **13** of the sample **10**.

[0093] In some embodiments, the position of the supporting portion **110** may be fixed and may serve as a reference for determining the position of other components. However, if necessary, the supporting portion **110** may be moved in the third direction D3 and drawn out directly from the accommodation space **415** of the sample arrangement portion **410**.

[0094] FIG. 8 shows the pressing state P3 of the sample **10** during the segmentation process of the sample **10**. In the pressing state P3, the pressing portion **210** moves toward the other surface **12** of the sample **10**, so that sample **10** may be pressed toward one surface **11** of the sample **10** (e.g., along third direction D3).

[0095] The pressing portion **210** may be provided as a pair and press the pair of pressing regions **14**. The pair of pressing units **210** may be moved together. When the other surface **12** of the sample **10** is pressed by the pair of pressing portion **210**, the bending force centered on the supporting portion **110** may act on the sample **10**. In some embodiments, the pressing force by the pressing portion **210** may act as a bending force for the sample **10**.

[0096] The supporting portion **110** may be positioned at the center between the pair of pressing units **210** with reference to the arrangement direction of the pair of the pressing units **210** (e.g., first direction D1). In other words, one region **13** of the sample **10** by the supporting portion **110** may be positioned in the center between the pair of pressing regions **14**.

[0097] Accordingly, the pressing force or the bending force acting on the sample **10** centered on the one region **13** by the supporting portion **110** uniformly acts on the sample **10**, thereby improving the formation accuracy of the cross-section.

[0098] FIG. 9 shows the indentation state P4 of the sample **10** during the segmentation process of the sample **10**. In the indentation state P4, the indentation portion **310** may engrave the indentation on the other surface **12** of the sample **10** while moving in the third direction D3 toward the other surface **12** of the sample **10**.

[0099] If the indentation occurs in the sample **10** where the bending force is generated by the pressing portion **210**, the splitting may occur while the stress is concentrated around the indentation. In some embodiments, the method of splitting the sample **10** by using the bending

force involves a relatively shallow depth of the indentation by the indentation portion **310** and may increase the formation accuracy of the cross-section compared to segmenting the entire sample **10** by using the indentation portion **310**. In other words, the cross-section may, in some embodiments, be as smooth as a mirror.

[0100] The segmentation region **15** of the sample **10** indented by the indentation portion **310** may be positioned between the pair of pressing regions **14** and may coincide with one region **13** of the supporting portion **110** described above. That is, the one region **13** and the segmentation region **15** may overlap along the third direction.

[0101] Through this, the uniformity of the pressing force/bending force acting on the sample **10** centered on the indentation mark may be improved, thereby forming the cross-section of the specimen more smoothly, according to some embodiments.

[0102] FIG. **10** shows the segmentation state P5 of the sample **10** during the segmentation process of the sample **10**. The sample **10** split in the above-described indentation state P4 may be segmented into two specimens and arranged on the sample arrangement portion **410**.

[0103] The sample arrangement portion **410** may move the specimens to an apparatus for the inspection, or the user may directly collect the specimens from the sample arrangement portion **410**. The specimen produced from the sample **10** may be used to evaluate the results of the process through a photography method of the cross-section.

[0104] FIG. **11** shows the supporting part **100** of the wafer sample segmentation apparatus **1**.

[0105] The supporting part **100** may include the supporting portion **110** described above and a supporting body **120**. The supporting body **120** may be arranged such that the supporting body **120** is spaced apart from the sample **10**. The supporting body **120** may be spaced apart from the sample **10** along the second direction D2.

[0106] The supporting portion **110** may be in the form of a bar/arm extending from the supporting body **120** toward the sample **10**. The supporting portion **110** may extend from the supporting body **120** and be positioned to face one surface **11** of the sample **10**.

[0107] The supporting portion **110** may include a contact portion that is in directly contact with one region **13** of the sample **10** and a reinforcing portion to reinforce the rigidity of the contact portion. The reinforcing portion may be positioned on the opposite side of the sample **10** with the contact portion as a reference, and may reinforce the rigidity of the contact portion to which the pressing force by the pressing portion **210** is transmitted.

[0108] The position and shape of the supporting body **120** may vary, and FIG. **11** shows, according to some embodiments, a block-shaped supporting body **120** positioned lower than the sample **10**.

[0109] A measuring part **150** may be provided in the supporting part **100**. The measuring part **150** may be configured to measure the pressing force acting on the sample **10** along the pressing direction of the pressing portion **210**.

[0110] The pressing force provided to the sample **10** is transmitted to the supporting portion **110** and the supporting body **120** by the pressing portion **210**. The measuring part **150** may be provided on the supporting body **120** and measure the pressing force transmitted to the supporting body **120** through the supporting portion **110**.

[0111] The measuring part **150** may be positioned such that the measuring part **150** is spaced apart from sample **10** along the pressing direction (e.g., the third direction D3) by the pressing portion **210** of the pressing body **220**. For example, the measuring part **150** may be provided at the end of the supporting body **120**, with the pressing direction as a reference, to support the supporting body **120** and to receive and/or facilitate the measurement of the pressing force acting on the sample **10**.

[0112] FIG. **11** shows the supporting body **120** arranged lower than the sample **10**, and the measuring part **150** that is coupled to the lower end of the supporting body **120** to support the supporting body **120**, according to some embodiments.

[0113] However, the position of the measuring part **150** is not necessarily limited as above, and the measuring part **150** may be built into the supporting body **120** or may be provided on the upper

side of the supporting body **120**. The measuring part **150** may be provided as a load cell type coupled to the supporting body **120**.

[0114] When manufacturing the specimen with the cross-section, the pressing force acting on the sample **10** is a relatively important factor in improving the smoothness of the cross-section.

[0115] Specifically, the smoothness of the specimen cross-section is a relatively important factor and may at least partially rely on a generation speed of the cross-section that is constant and appropriate, according to some embodiments.

[0116] If the bending force of the sample **10** is too large, a damage may occur in the cross-section during the splitting process of the sample **10**, and if the bending force is too small, the depth of the indentation mark caused by the indentation portion **310** may become deeper and the damage of the cross-section may occur due to the contact with the indentation portion **310**.

[0117] In addition, if the bending force of the sample **10** fluctuates unstably, it may be difficult to obtain the uniform cross-section flatness, which may be detrimental to the cross-section inspection.

[0118] As the wafer sample segmentation apparatus **1** according to some embodiments is provided with the measuring part **150** in the supporting part **100** to support the sample **10**, during the segmentation process of the sample **10**, the pressing force, or the bending force acting on the sample **10**, may be adjusted to the constant and appropriate level to produce the specimen.

[0119] For example, in some embodiments, if a measurement value of the measuring part **150** in the pressing state **P3** in which the pressing portion **210** presses the sample **10** is within a pressing reference range, the indentation portion **310** may be moved toward the sample **10** to indent the segmentation region **15**.

[0120] The pressing reference range is a range of the pressing force that is theoretically and/or experimentally determined in advance to improve the formation accuracy of the cross-section of the specimen. By performing the indentation process when the pressing force acting on the sample **10** through the pressing portion **210** is within the pressing reference range, the specimen of the cross-section with high formation accuracy may be produced.

[0121] The control of the pressing force acting on the sample **10** may be relatively important, not only in the pressing state **P3**, but also in the indentation state **P4**. For example, during the indentation process of the sample **10**, the pressing force acting on the sample **10** may be increased by the indentation portion **310** in addition to the pressing portion **210**.

[0122] As mentioned above, since the generation speed of the cross-section is a relatively important factor in improving the formation accuracy of the cross-section, and it may be advantageous that the speed at which this cross-section occurs is maintained during the indentation process, some embodiments may keep the pressing force acting on the sample **10** appropriate and constant through the measurement value measured by the measuring part **150** even during the indentation process.

[0123] For example, in the indentation process in which the pressing portion **210** presses the sample **10** and the indentation portion **310** presses and indents the sample **10**, the indentation portion **310** may be moved to indent the segmentation region **15** so that the measurement value of the measuring part **150** falls within the indentation reference range.

[0124] The indentation reference range may be a range of the pressing force during the indentation process that is theoretically and/or experimentally determined in advance to improve the formation accuracy of the cross-section of the specimen. In some embodiments, the indentation reference range may be the same as the aforementioned pressing reference range, or at least a portion of the range may be defined within the pressing reference range, or at least a portion of the pressing reference range may be defined within the indentation reference range.

[0125] FIG. **12** shows the pressing part **200** of the wafer sample segmentation apparatus **1**, according to some embodiments. The pressing part **200** may include a pressing body **220** in addition to the pressing portion **210** described above.

[0126] The pressing body **220** may be arranged to be spaced apart from the sample **10**, and the pair

of pressing portions **210** may be protruded from the pressing body **220** to face the other surface **12** of the sample **10**.

[0127] The pair of pressing portions **210** may be arranged along the first direction **D1**, and the pressing bodies **220** may be spaced apart in the second direction **D2** with respect to the sample **10**. The pressing portion **210** may extend along the second direction **D2** from the pressing body **220** toward the sample **10** and be positioned to face the other surface **12** of the sample **10**.

[0128] The pressing body **220** may be provided in the form of a block from which the pressing portion **210** is protruded, and the pair of pressing portions **210** may be protruded together from one pressing body **220**.

[0129] Meanwhile, the pressing body **220** may be arranged to be movable along the pressing direction and the arrangement direction. That is, the pressing body **220** may be provided to be movable along the first direction **D1** and the third direction **D3**, and the pressing portion **210** may be moved by the movement of the pressing body **220** to press the sample **10** or the position thereof may be adjusted.

[0130] The pressing part **200** may include a pressing moving part **230** for moving the pressing body **220**. The pressing moving part **230** may be provided in various types and shapes that may move the pressing body **220**.

[0131] FIG. **12** shows, as an example, the pressing moving part **230** in the form of a block to which the pressing body **220** is movably coupled. The pressing body **220** may be capable of moving in the first direction **D1** and the third direction **D3** while coupled to the pressing moving part **230**, and the pressing moving part **230** may include an actuator for the pressing portion **210** that generates the moving force of the pressing body **220**.

[0132] FIG. **13** shows the indentation part **300** of the wafer sample segmentation apparatus **1**, according to some embodiments. The indentation part **300** may include an indentation body **320** along with the indentation portion **310** described above.

[0133] The indentation body **320** may be arranged to be spaced apart from sample **10**, and the indentation portion **310** may be provided on the indentation body **320** to indent the sample **10** by being protruded from the indentation body **320** toward the other surface **12** of the sample **10**.

[0134] The position of the indentation body **320** for the sample **10** may vary. For example, as shown in FIG. **13**, the indentation body **320** may be arranged to be spaced apart in the third direction **D3** approximately for the sample **10**, and the indentation portion **310** may be provided in a protruded form from the indentation body **320** toward the other surface **12** of the sample **10**.

[0135] The indentation body **320** may be provided with a plurality of indentation portions **310**. The indentation portion **310** is in direct contact with the sample **10** and indents the sample **10**, and, as a durability of each indentation portion **310** progresses through the repeated indentation processes, the indentation portion **310** that has undergone a predetermined number of the indentation processes may be replaced with a new indentation portion **310**.

[0136] For this purpose, the indentation body **320** may be pre-equipped with a plurality of indentation portions **310**, and the indentation body **320** may perform the indentation process by replacing the indentation portion **310** currently in use with another indentation portion **310** after a predetermined number of indentation processes have been performed.

[0137] There may be various ways in which the indentation body **320** is provided with a plurality of indentation portions **310** to enable replacement. For example, as shown in FIG. **13**, the indentation body **320** may have an approximately circular cross-section shape and be rotatable about a rotation axis parallel to the second direction **D2**, and the plurality of indentation portions **310** may be arranged along the circumferential direction of the indentation body **320**.

[0138] When the use number of the indentation portion **310** facing the current sample **10** exceeds a predetermined number, the indentation body **320** may be rotated to arrange another indentation portion **310** adjacent to the indentation portion **310** currently in use to face the sample **10**.

[0139] The indentation part **300** may be prepared so that the indentation portion **310** may move in a

direction parallel to one surface **11** of the sample **10**. The indentation portion **310** may be moved in at least one direction among the first direction **D1**, the second direction **D2**, and the third direction **D3**.

[0140] The indentation body **320** may be provided to move the indentation portion **310**, and the movement of the indentation portion **310** may be accomplished by the movement of the indentation body **320**.

[0141] The indentation part **300** may include an indentation moving part **330** for moving the position of the indentation body **320**. The indentation body **320** may be movably coupled to the indentation moving part **330**. The indentation moving part **330** may include an indentation actuator to generate the moving force of the indentation body **320**.

[0142] Furthermore, the indentation portion **310** may be arranged to be rotatable around the rotation axes parallel to each of the first direction **D1** parallel to one surface **11** of the sample **10**, the second direction **D2** parallel to one surface **11** of the sample **10** and orthogonal to the first direction **D1**, and the third direction penetrating the sample **10**.

[0143] Since the indentation portion **310** may be rotatably provided on the indentation body **320** and/or the indentation body **320** may be rotated by the indentation moving part **330**, rotational movement of the indentation portion **310** may occur.

[0144] The indentation portion **310** may be rotated around the rotation axis parallel to the first direction **D1** so that the blade of the indentation portion **310** may be aligned to be parallel to the other surface **12** of the sample **10**. Specifically, the end of the indentation portion **310** facing the sample **10** may be defined as a blade which may make contact with the sample **10** and indent the sample **10**.

[0145] The blade may extend along the second direction **D2** to be in contact with the sample **10**. However, if the blade is not parallel to the other surface **12** of the sample **10**, some of the segmentation region **15** of the sample **10** may be indented first along the second direction **D2**, and the rest may be indented later, which may deteriorate the formation accuracy for the entire cross-section of the specimen.

[0146] Therefore, the indentation portion **310** may be rotated around the rotation axis parallel to the first direction **D1** and aligned in parallel with the other surface **12** of the specimen such that the segmentation region **15** may be simultaneously indented as a whole.

[0147] The indentation portion **310** may be rotated around the rotation axis parallel to the second direction **D2**, so that the indentation direction of the sample **10** of the blade may be precisely matched to the third direction **D3**. Additionally, as the indentation portion **310** is rotated around the rotation axis parallel to the third direction **D3**, the blade direction **D5** (see FIG. 17), where the blade extends, and the crystal direction **D4** (see FIG. 17) present in the sample **10** may be aligned to be matched.

[0148] FIG. 14 shows the sample arrangement part **400** of the wafer sample segmentation apparatus **1**, according to some embodiments. The sample arrangement part **400** may include the sample arrangement portion **410** described above and a sample arrangement body **420**.

[0149] The sample arrangement portion **410** may provide a work space where the sample **10** is arranged and the segmentation process is performed, and the sample arrangement portion **410** may be provided on the sample arrangement body **420** arranged such that the sample arrangement portion **410** is spaced apart from the sample **10** and supported by the sample arrangement body **420**.

[0150] The sample arrangement body **420** is arranged to be spaced apart from sample **10**, and the shape and position thereof may vary. For example, the sample arrangement body **420** may be positioned on the opposite side of the above-described pressing part **200** and indentation part **300** centered on the sample **10**, for appropriate utilization of space.

[0151] The sample arrangement portion **410** may be moved along the pressing direction and may be provided to rotate the sample **10** about the rotation axis parallel to the pressing direction.

[0152] In some embodiments, the sample arrangement portion **410** may be moved along the third direction **D3** or rotated about the rotation axis parallel to the third direction **D3** with respect to the sample arrangement body **420**. In some embodiments, the sample arrangement body **420** may be directly moved and rotated.

[0153] For example, the sample arrangement part **400** may include a sample arrangement moving part **430**. The sample arrangement moving part **430** may be coupled to and may move the sample arrangement body **420**. The sample arrangement moving part **430** may include an actuator for the sample **10** arrangement to provide the moving force for the sample arrangement body **420**.

[0154] FIG. **15** shows a camera provided in the wafer sample segmentation apparatus **1**, according to some embodiments.

[0155] The process of segmenting the sample **10** to produce the specimen having the cross-section involves relatively precise control, and for this purpose, at least one or more cameras may be installed to identify and align the position relationships between components.

[0156] In an embodiment, the wafer sample segmentation apparatus **1** may include a first camera **500**, and the first camera **500** may be spaced apart from the sample **10** along the protrusion direction of the pressing portion **210** to image the pressing portion **210** and the supporting portion **110**.

[0157] The first camera **500** may be positioned to be spaced apart from the sample **10** in the second direction **D2**, and may be positioned on the opposite side of the supporting portion **110** and the pressing portion **210** with respect to the sample **10**. The first camera **500** may photograph at least one of the supporting portion **110**, the pressing portion **210**, and the indentation portion **310**.

[0158] The wafer sample segmentation apparatus may include a second camera **600**, and the second camera **600** may be spaced apart from the sample **10** along the pressing direction and photograph the crystal direction **D4** of the sample **10**.

[0159] The second camera **600** may be arranged away from sample **10** along the third direction **D3**, and may be positioned on the opposite side of the sample arrangement portion **410** and the supporting portion **110** with the sample **10** as the center. The second camera **600** may be positioned above the sample **10**. The second camera **600** may photograph at least one of the other surface **12** of the sample **10** and the indentation portion **310**.

[0160] FIG. **16** shows the pressing portion **210** and the indentation portion **310** moving along the first direction **D1** in the wafer sample segmentation apparatus **1**, according to some embodiments.

[0161] In some embodiments, the pair of pressing portion **210** may be arranged to be movable along the arrangement direction of the pair of pressing portion **210**, for example the first direction **D1**, as described above.

[0162] As mentioned above, in order to provide the uniform pressing force to the sample **10**, it may be advantageous that the supporting region of the sample **10** by the supporting portion **110** is positioned in the center between the pair of pressing regions **14** defined by the pair of pressing portions **210**.

[0163] Accordingly, in some embodiments, the pair of pressing portions **210** may be moved along the arrangement direction such that the supporting portion **110** may be positioned at the center between the pair of pressing portions **210**, and the position relationship between the pressing portion **210** and the supporting portion **110** may be confirmed directly by the user or through the photographic information of the first camera **500** described above.

[0164] Also, as mentioned above, in order to improve the formation accuracy of the cross-section of the sample **10**, it may be advantageous for the segmentation region **15** and the support region to coincide. In some embodiments, the segmentation region **15** and the support region may overlap each other in the third direction, and accordingly, the indentation portion **310** and the supporting portion **110** may be aligned along the pressing direction so that the segmentation region **15** and one region **13** by the supporting portion **110** may be matched.

[0165] For example, the indentation portion **310** may be aligned with the supporting portion **110**

along the third direction D3 by adjusting the position thereof along the first direction D1 with the position of the supporting portion 110 as a reference. The position of the indentation portion 310 may be checked directly by the user or through the shooting information of the first camera 500. [0166] FIG. 17 shows a process of aligning the crystal direction D4 of the sample 10 and the blade direction D5 of the indentation portion 310, according to some embodiments.

[0167] The sample 10 provided from the wafer has the crystal direction D4 like the wafer, and if the crystal direction D4 and the blade direction D5 of the indentation portion 310 are aligned in parallel, it may be advantageous to improve the smoothness of the cross-section.

[0168] As described above, in the wafer sample segmentation apparatus 1 according to some embodiments, at least one of the indentation portion 310 and the sample arrangement portion 410 may be rotated around the rotation axis parallel to the third direction D3.

[0169] The indentation portion 310 may adjust the blade direction D5 through the rotation, and the sample arrangement portion 410 may adjust the crystal direction D4 of the sample 10 through the rotation. The alignment process of the blade direction D5 and the crystal direction D4 may be checked directly by the user with the naked eye or based on shooting information from the second camera 600.

[0170] FIG. 18 shows a block diagram showing a control relationship by a controller 50 of the wafer sample segmentation apparatus 1 in some embodiments.

[0171] In some embodiments, a controller 50 may control the movement of at least one of the pressing portion 210 and the indentation portion 310. The controller 50 may receive the measurement value from the measuring part 150.

[0172] Information input group G1 for the controller 50 may comprise the above-described first camera 500 and second camera 600. The components of the information input group G1 may contribute to control activities of the controller 50 by providing information to the controller 50.

[0173] Controlled group G2 for the controller 50 may comprise the above-described pressing part 200, indentation part 300, and sample arrangement part 400. The movements of the components of the controlled group G2 may be controlled according to the instructions of the controller 50. The pressing part 200 may be defined as including the aforementioned measuring part 150.

[0174] For the first camera 500, the pressing part 200, and the indentation part 300, the control of the pressing part 200 and the indentation part 300 may be performed based on the shooting information of the first camera 500 as the first group G3 for the controller 50.

[0175] For the second camera 600 and the sample arrangement part 400, the control of the sample arrangement part 400 may be performed based on the shooting information of the second camera 600 as the second group G4 for the controller 50. As described above, when the indentation portion 310 rotates around the third direction D3, the indentation portion 310 may also be defined as being included in the second group G4.

[0176] The controller 50 may align the position relationship between the supporting portion 110, the pressing portion 210, and the indentation portion 310 according to the first direction D1 based on the shooting information of the first camera 500. The controller 50 may adjust the mutual position relationship by moving the pressing portion 210 and the indentation portion 310 along the first direction D1 around the supporting portion 110. If necessary, a position reference other than supporting portion 110 may be selected.

[0177] The controller 50 may align the crystal direction D4 of the sample 10 and the blade direction D5 of the indentation portion 310 based on the captured information of the second camera 600. The controller 50 may rotate the sample arrangement portion 410 or rotate the indentation portion 310 to adjust the rotation position relationship between them.

[0178] As described above, the controller 50 may control the movement of at least one of the pressing portion 210 and the indentation portion 310 based on the measurement value of the measuring part 150.

[0179] The controller 50 may receive the measurement value of the measuring part 150 and may

determine whether the measurement value belongs within the pressing reference range or the indentation reference range, thereby controlling at least one of the movement of the pressing portion **210** or the movement of the indentation portion **310**.

[0180] While the present disclosure has been particularly shown and described with reference to specific embodiments thereof, it will be understood by those skilled in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the disclosure as defined by the following claims.

Claims

1. A wafer sample segmentation apparatus, comprising: a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part facing the other surface facing one surface of the sample and capable of pressing a pair of pressing regions positioned on both sides of the one region of the sample by being moved toward the sample; an indentation part facing the other surface of the sample and capable of moving toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample; and a measuring part capable of measuring a pressing force acting on the sample along the pressing direction of the pressing part.
2. The wafer sample segmentation apparatus of claim 1, wherein: in a pressing state where the pressing part is pressing the sample, if the measurement value of the measuring part falls within the pressing reference range, the indentation part is moved toward the sample to indent the segmentation region.
3. The wafer sample segmentation apparatus of claim 1, further comprising: a supporting body arranged away from the sample, the supporting part is protruded from the supporting body to face one surface of the sample, and the measuring part is provided on the supporting body and is capable of measuring the pressing force transmitted to the supporting body through the supporting part.
4. The wafer sample segmentation apparatus of claim 3, wherein: the measuring part is provided at the end of the supporting body, with reference to the pressing direction, to support the supporting body.
5. The wafer sample segmentation apparatus of claim 3, wherein: the measuring part is provided as a load cell type coupled to the supporting body.
6. The wafer sample segmentation apparatus of claim 1, wherein: the pressing part is provided as a pair and is capable of pressing the pair of pressing regions, and the supporting part is positioned in the center, with reference to the arrangement direction of the pair of pressing parts, between the pair of pressing parts.
7. The wafer sample segmentation apparatus of claim 6, further comprising: a pressing body arranged away from the sample, wherein: the pair of pressing parts protruding from the pressing body to face the other surface of the sample, and the pressing body is movable along the pressing direction and the arrangement direction.
8. The wafer sample segmentation apparatus of claim 7, wherein: a first camera spaced apart from the sample along a protrusion direction of the pressing part and capable of photographing the pressing part and the supporting part.
9. The wafer sample segmentation apparatus of claim 1, wherein: the indentation part and the supporting part are aligned along the pressing direction so that the segmentation region and the one region of the sample overlap each other along the pressing direction.
10. The wafer sample segmentation apparatus of claim 9, further comprising: an indentation body arranged away from the sample, wherein the indentation part is protruded from the indentation body toward the other surface of the sample, and the indentation body is provided so that the indentation part is movable in a direction parallel to one surface of the sample.

- 11.** The wafer sample segmentation apparatus of claim 1, wherein: the indentation part is provided to be rotatable about a rotation axis respectively parallel to a first direction parallel to the one surface of the sample, a second direction parallel to the one surface of the sample and vertical to the first direction, and a third direction penetrating the sample.
 - 12.** The wafer sample segmentation apparatus of claim 1, further comprising: a sample arrangement part where the sample and the supporting part are positioned, wherein the sample arrangement part is arranged to rotate the sample about a rotation axis parallel to the pressing direction.
 - 13.** The wafer sample segmentation apparatus of claim 12, further comprising: a second camera spaced away from the sample along the pressing direction and capable of photographing the crystal direction of the sample.
 - 14.** The wafer sample segmentation apparatus of claim 1, further comprising: a controller that receives the measurement value from the measuring part and is capable of controlling the movement of at least one of the pressing part or the indentation part.
 - 15.** A wafer sample segmentation apparatus comprising: a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part facing the other surface opposite to the one surface of the sample and capable of moving toward the sample to press the pressing region of the sample; an indentation part facing the other surface of the sample and capable of moving toward the sample to indent the segmentation region of the sample; and a measuring part capable of measuring a measurement value acting on the sample along the pressing direction of the pressing part, wherein: if the measurement value of the measuring part falls within a pressing reference range, the indentation part is capable of moving toward the sample to indent the segmentation region.
 - 16.** The wafer sample segmentation apparatus of claim 15, wherein: in an indentation process, where the pressing part is capable of pressing the sample and the indentation part is capable of pressing and indenting the sample, the indentation part is capable of moving to indent the segmentation region so that the measurement value of the measuring part falls within an indentation reference range.
 - 17.** The wafer sample segmentation apparatus of claim 16, wherein: a controller receiving the measurement value of the measuring part and determining whether the measurement value falls within the pressing reference range or the indentation reference range such that at least one of the movement of the pressing part or the movement of the indentation part is controlled.
 - 18.** A wafer sample segmentation apparatus comprising: a supporting part supporting one region of a sample of a wafer while facing one surface of the sample; a pressing part provided as a pair facing the other surface opposite to the one surface of the sample, and capable of moving toward the sample to press a pair of pressing regions positioned on both sides of the one region in the sample; and an indentation part facing the other surface of the sample and capable of moving toward the sample to indent a segmentation region positioned between the pair of pressing regions in the sample, wherein the supporting part is positioned between the pair of pressing parts with reference to an arrangement direction of the pair of pressing parts, and the pair of pressing parts is movable along the arrangement direction.
 - 19.** The wafer sample segmentation apparatus of claim 18, wherein: the pair of pressing parts are capable of moving along the arrangement direction so that the supporting part is positioned centrally between the pair of pressing parts.
 - 20.** The wafer sample segmentation apparatus of claim 18, further comprising: a pressing body arranged spaced apart from the sample, wherein: the pair of pressing parts is protruded from the pressing body to face the other surface of the sample, and the pressing body is movable along the pressing direction and the arrangement direction.
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